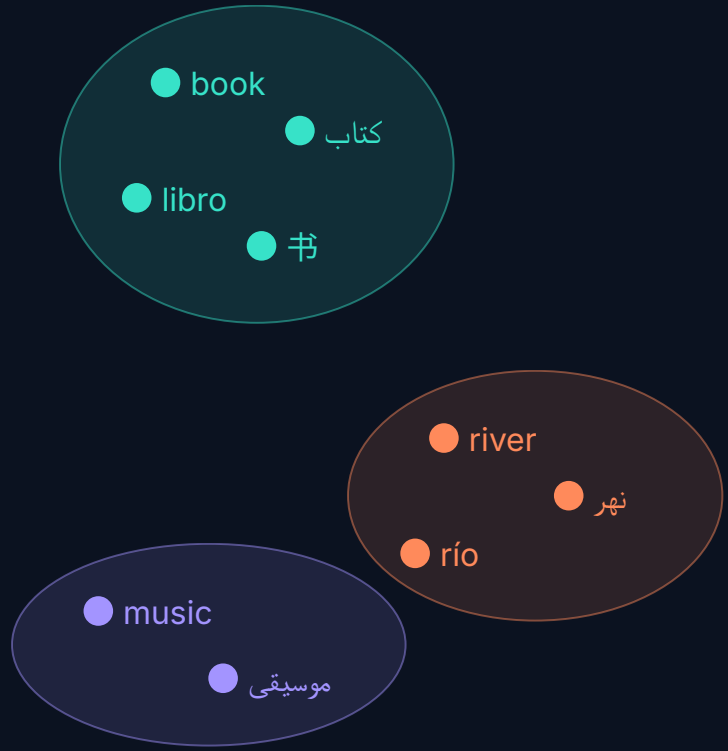
CS 410 TECH REVIEW

Searching Across Languages

Cross-lingual and multilingual information retrieval

search · بحث · búsqueda · 搜索 · recherche



Zaina Khalil · Fall 2026

INTRODUCTION

Why I picked this topic

لغة

I live in two languages

I'm a heritage Arabic speaker and was a TA for Arabic courses at UIC, so I notice how much information only exists in one language.

Linguistics meets IR

My independent study on bilingual cognitive processing made me curious how search systems handle the same problem.

RAG is going global

LLM apps retrieve before they answer. If retrieval only works well in English, everyone else gets worse answers.

BACKGROUND

The answer isn't always in your language

50%+

of digital content is produced in languages other than English

Statista (2025), cited in Ki et al. 2025

QUERY · ENGLISH

how do vaccines work

BEST DOCUMENT · ARABIC

اللقاحات تدرّب جهاز المناعة على التعرف على الفيروسات

"Vaccines train the immune system to recognize viruses"

Zero shared words, so BM25 scores it 0. The meaning matches; the vocabulary doesn't.

Two flavors of the problem

idioma

CLIR

Cross-language IR

Query in one language, documents in another.

EXAMPLE

English query → Persian news

MLIR

Multilingual IR

Query in one language, documents in many. Return one merged ranked list.

EXAMPLE

English query → Chinese + Persian + Russian news

Why MLIR is harder: scores computed for documents in different languages usually aren't comparable, so merging the lists is its own problem (TREC NeuCLIR 2023).

HOW IT WORKS

Three ways across the language barrier

翻译

1 Translate the query

- English query
- MT → Persian
- BM25 on Persian docs

Cheap. Queries are short, so one bad translation hurts everything.

2 Translate the documents

- Persian docs
- MT → English (offline)
- BM25 in English

MT gets more context, but translating a whole collection is costly.

3 Shared embedding space

- Query + docs
- one multilingual encoder
- nearest neighbors

No MT at search time. Now the default for neural CLIR.

HOW IT WORKS

One vector space for 100+ languages

向量

Contrastive pre-training

- Collect ~1 billion related multilingual text pairs (title + passage, question + answer, translations)
- Pull each pair's vectors together, push the rest of the batch apart (InfoNCE loss)
- Fine-tune on labeled retrieval data such as MS MARCO and MIRACL

Recipe behind multilingual-E5 (Wang et al. 2024)

Multilingual knowledge distillation

English teacher	Hello World	`[0.8, -0.2, 0.3]`
Student	Hello World	`[0.7, -0.1, 0.3]`
Student	Hallo Welt	`[0.9, -0.2, 0.4]`

Train the student so a sentence and its translation both land on the teacher's English vector (MSE loss). Vectors are illustrative.

Reimers & Gurevych 2020

TOOL SPOTLIGHT

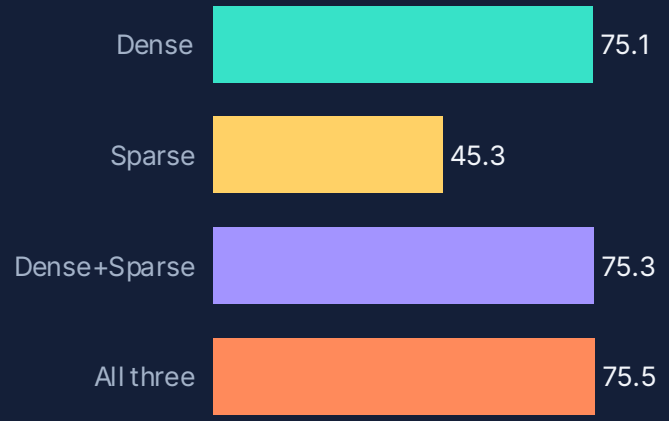
BGE-M3: one model, three kinds of retrieval

Dense — One vector per text (the [CLS] token). Fast nearest-neighbor search.

Sparse — A learned weight for each token. Like a learned version of TF-IDF / BM25 term weighting.

Multi-vector — One vector per token, scored ColBERT-style. Finer matching, more storage.

Cross-lingual recall (MKQA, 25 languages)



Sparse alone struggles across languages: few shared terms. Source: Chen et al. 2024

HOW IT WORKS

Where does CLIR training data come from?

The big labeled sets like MS MARCO are English only. ColBERT-X (Nair et al., ECIR 2022) tries two fixes:

Zero-shot

Train on English MS MARCO only. Rely on the multilingual XLM-R encoder to carry it to other languages.

Translate-train

Machine-translate the MS MARCO passages into the target language, keep the English queries, then train.

Result: ColBERT-X beat traditional lexical CLIR baselines. Translate-train won on 5 of 7 test collections, but zero-shot was better on Chinese and Russian.
Tested with English queries on HC4 (Chinese, Persian) and CLEF (French, German, Italian, Russian, Spanish).

How we measure it

Benchmark	Setup	Why it matters
MIRACL	18 languages, ~77k queries over Wikipedia, judged by native speakers (TACL 2023)	The standard multilingual retrieval benchmark
TREC NeuCLIR	2022–2024. English queries over Chinese, Persian and Russian news; Chinese technical abstracts; report generation	Translation and neural CLIR tested side by side
MMTEB	500+ tasks across 250+ languages, community built (ICLR 2025)	Best place to compare embedding models
CIRAL	English queries over Hausa, Somali, Swahili and Yoruba passages (FIRE 2023)	Shows how low-resource languages fare

Metric you'll see most: nDCG@10, which rewards putting the most relevant documents first.

APPLICATIONS

Where it's used

بحث

Multilingual RAG

Assistants that retrieve evidence in any language before they answer

Foreign-language news search

English queries over Chinese, Persian and Russian news, as in NeuCLIR

Research literature

Finding non-English papers, like NeuCLIR's Chinese technical abstracts task

Support and FAQ search

One help-center index that answers users in whatever language they write

TRY IT YOURSELF

Cross-lingual search in a few lines

0.82

```
from sentence_transformers import SentenceTransformer, util

model = SentenceTransformer("intfloat/multilingual-e5-base")

query = "query: how do vaccines work"
docs  = ["passage: " + d for d in (ar_doc, es_doc, en_doc)]

q = model.encode(query, normalize_embeddings=True)
D = model.encode(docs,  normalize_embeddings=True)
print(util.cos_sim(q, D))
```

ar_doc scored highest (0.82) with zero words in common with the query. E5 scores bunch up around 0.7 to 0.9, so the ranking is what matters.

ar_doc **score 0.822**

اللقاحات تدرّب جهاز المناعة على التعرف على الفيروسات

Vaccines train the immune system to recognize viruses

es_doc **score 0.723**

La paella lleva arroz y azafrán

Paella has rice and saffron

en_doc **score 0.715**

The stock market fell on Monday

COMPARISON

Translation vs. shared embeddings

	Translate + BM25	Shared embeddings
Search-time cost	MT on every query, or the whole corpus offline	One encoder pass; one index for all languages
Explainability	Easy: you can read the translated terms	Hard: closeness in a learned vector space
Exact names and terms	Strong when MT gets them right	Can blur exact matches (hybrids add sparse)
Low-resource languages	Limited by MT quality	Limited by pre-training data
Ranking across languages	Scores often not comparable across languages	One score space, but can favor some languages

LIMITATIONS

The open problem: language bias

Linguistic nepotism

Ki et al. 2025 · UMD & Johns Hopkins

Tested 8 languages (including Arabic) and 6 open LLMs in multilingual RAG. Models preferred to cite English documents, sometimes even irrelevant ones. Swahili saw the biggest drop: -23.9% accuracy on average.

Or is it the data?

Park et al. 2026 · DELTA

Argues much of the English preference comes from where answers exist: English sources hold more of the gold passages. Retrievers mainly favor documents in the query's own language.

Takeaway: check which languages your top-k results actually come from.

MY THOUGHTS

What I took away

فكرة

1

Start with a shared embedding model

For a course project I'd begin with multilingual-E5 or BGE-M3 plus a reranker. A few lines of code already cover 100+ languages.

2

Benchmarks don't match how people type

MIRACL's Arabic data comes from Wikipedia, which is mostly formal written Arabic. Real users often search in dialect or Arabizi, and I'd like evaluation that reflects that.

3

Fairness is a retrieval problem too

Language bias starts before the LLM writes anything. Checking the language mix of retrieved results should be a normal evaluation step.

RESOURCES

Where I'd start reading

اقرأ

Galuščáková, Oard & Nair: CLIR chapter
History and vocabulary of the field
arxiv.org/abs/2111.05988

MIRACL (TACL 2023)
The main multilingual retrieval benchmark
miracl.ai

TREC NeuCLIR track
CLIR approaches compared on the same data
neuclir.github.io

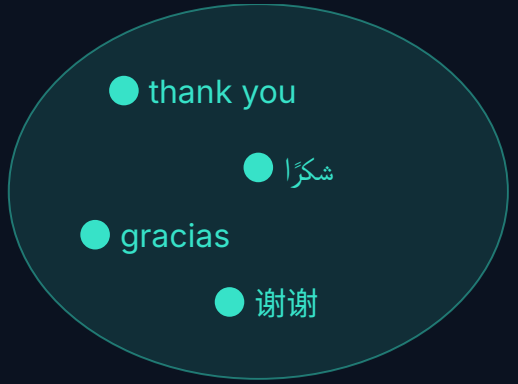
Multilingual-E5 and BGE-M3 papers
How today's open models are trained
arxiv.org/abs/2402.05672 · 2402.03216

Sentence Transformers docs
Hands-on code for multilingual models
sbert.net

Linguistic Nepotism (Ki et al.)
Language bias in multilingual RAG
arxiv.org/abs/2509.13930

Thank you

Questions and resources



TEXTDATA COMMUNITY

Cross-Lingual & Multilingual Information Retrieval

textdata.org/community/6af11ce3a8567d79c2f88579